

What is a Server?

A **Server** is a **computer or software system** that **provides services, resources, or data** to other computers called **clients** over a **network**.

Simple Definition:

A server **serves requests**, and a client **requests services**.

Example:

- Your laptop → **Client**
- Google's system hosting youtube.com → **Server**

Why Servers Are Needed

Servers help in:

- Sharing files and data
- Hosting websites and applications
- Managing emails
- Authenticating users
- Providing security and backups

How Server Works (Basic Flow)

1. Client sends a request (browser, app, device)
2. Server receives the request
3. Server processes it
4. Server sends a response back to the client

Example:

When you open www.google.com

- Browser sends request
- Google server responds with webpage data

Types of Servers (With Explanation)

File Server

Purpose:

Stores and shares files over a network.

Example:

- Office shared folder

- College lab notes server

Protocols Used:

- SMB (Windows)
- NFS (Linux)

 Use Case:

Centralized storage for users.

2 Web Server**Purpose:**

Hosts websites and delivers web pages.

Examples:

- Apache
- Nginx
- Microsoft IIS

Protocols Used:

- HTTP / HTTPS

 Use Case:

Hosting websites like blogs, company portals.

3 Application Server**Purpose:**

Runs business logic and applications.

Examples:

- Tomcat
- JBoss
- WebLogic

 Use Case:

Banking apps, ERP systems, college portals.

4 Database Server**Purpose:**

Stores, manages, and retrieves databases.

Examples:

- MySQL
- Oracle
- SQL Server

- PostgreSQL

 Use Case:

User records, transaction data, logs.

5 Mail Server

Purpose:

Sends, receives, and stores emails.

Protocols Used:

- SMTP (sending)
- POP3 / IMAP (receiving)

 Examples:

- Gmail
- Outlook mail servers

6 DNS Server (Domain Name System)

Purpose:

Converts **domain names** into **IP addresses**.

 Example:

www.google.com → 142.250.195.36

 Importance:

Without DNS, we must remember IP addresses.

7 Authentication Server

Purpose:

Verifies user identity and permissions.

Examples:

- Active Directory
- LDAP

 Use Case:

Login authentication in offices and colleges.

8 Proxy Server

Purpose:

Acts as a **middleman** between client and internet.

Benefits:

- Security
- Anonymity
- Caching
- Content filtering

Use Case:

Corporate networks, schools.

9 Print Server

Purpose:

Manages multiple printers in a network.

Use Case:

Offices and colleges with centralized printing.

10 Backup Server

Purpose:

Stores backup copies of data.

Use Case:

Disaster recovery, ransomware protection.

1 Virtual Server

Purpose:

Software-based server running on physical hardware.

Examples:

- VMware
- Hyper-V
- VirtualBox

Use Case:

Cost-saving and cloud environments.

2 Cloud Server

Purpose:

Server hosted on cloud infrastructure.

Examples:

- AWS EC2
- Microsoft Azure

- Google Cloud



Use Case:

Scalable applications, startups, enterprises.

Types of Servers Based on Hardware

Type	Description
Tower Server	Looks like a desktop
Rack Server	Mounted in server racks
Blade Server	High-density compact servers

Real-Life Example (Easy to Explain)



WhatsApp Example:

- App Server → Handles messages
- Database Server → Stores chats
- Authentication Server → Login
- Backup Server → Chat backups